

Figure 12: Cartoon of the modified boundary condition where scrambling in the environment, as indicated by the green region, can only occur up to the position of the impurity, the dashed line.

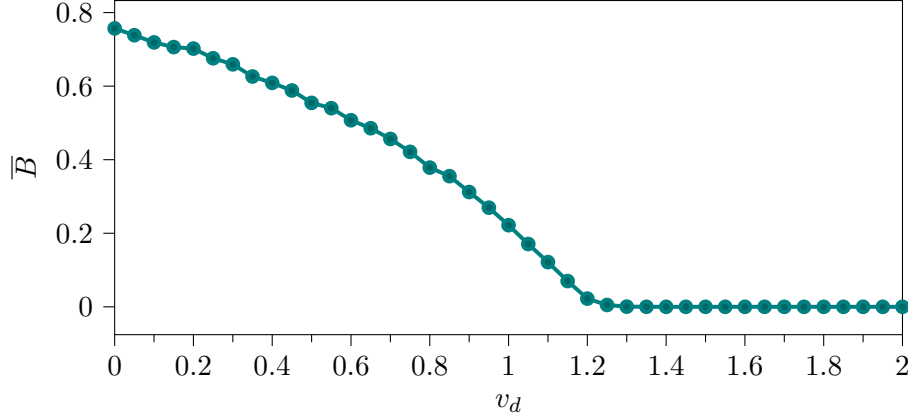


Figure 13: Backflow order parameter for the alternate circuit on variation of impurity drift velocity. Here we set $t_1 = 100$, $t_2 = 1000$ and $p = 0.1$. This is the result of an average over 10^4 disorder realizations.

at the impurity position which is instead given by

$$P_1(v_d, t) = \frac{\exp\left(\frac{-v_d^2 t}{4D}\right)}{\sqrt{\pi D t} \left[1 + \operatorname{erf}\left(\frac{v_d \sqrt{t}}{2\sqrt{D}}\right)\right]}. \quad (16)$$

This in turn satisfies a distinct scaling relation:

$$P_1(v, t) = \sqrt{t} F_1(v\sqrt{t}), \quad (17)$$

and we see this manifest as well in the scaling of the backflow order parameter in Figure 14